

puremacro: browser-runnable empirical macroeconomics in pure Python

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Summary

`puremacro` is a Python library for empirical macroeconomics and quantitative macro modeling, implemented entirely on the scientific-Python core — NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020), pandas, and Matplotlib — with no compiled or specialized dependencies at runtime. Because it avoids heavier libraries, it executes unchanged in the browser through Pyodide (The Pyodide development team 2024), on a tablet, or in a Jupyter-Lite notebook, as readily as on a workstation. The library spans the methods an applied macroeconomist reaches for: reduced-form and structural vector autoregressions (Sims 1980; Blanchard and Quah 1989), local projections (Jordà 2005), ARCH/GARCH volatility models (Engle 1982; Bollerslev 1986), heteroskedasticity- and autocorrelation-consistent inference (Newey and West 1987), dynamic-panel and weak-instrument estimators, difference-in-differences and synthetic control, value-function iteration and heterogeneous-agent equilibria (Aiyagari 1994), linear DSGE solution (Smets and Wouters 2007), and text-based uncertainty indices (Baker et al. 2016).

Statement of need

Macroeconometrics and computational macro are usually taught and practiced with tools that demand a substantial local installation — the `statsmodels` (Seabold and Perktold 2010), `arch`, and `linearmodels` stack — or commercial software, such as MATLAB for Dynare and most heterogeneous-agent toolkits. That barrier is invisible to well-resourced users and decisive for everyone else: students on tablets, instructors in low-bandwidth classrooms, and researchers without budgets or administrative rights. `puremacro` removes it. Because the library is pure scientific-Python, a reader can open a browser and reproduce a structural VAR, a local-projection impulse response, or an Aiyagari equilibrium at zero cost, with nothing to install.

Breadth alone is not trustworthy, so `puremacro` ships its own evidence. A built-

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Figure 1: Validation coverage: 73 cases across 13 subsystems, each checked against an independent reference.

in **validation gallery** contains 73 cases that continuously check the library’s estimators against independent references — statsmodels (Seabold and Perktold 2010), **arch**, and **linearmodels** where a canonical implementation exists, and closed-form solutions, published numbers, or internal cross-method identities elsewhere — across thirteen subsystems (Figure 1). A companion **replication gallery** reproduces published results from heterogeneous-agent and DSGE models. Both are wired into the CI workflows shipped with the package and re-run in a single line, `puremacro.validation.scorecard()`, including in the browser, so a user can verify the library on the same machine that runs it. No existing macro-econometrics package, to our knowledge, combines this breadth with browser portability and a continuously verified, self-contained correctness record.

Features

- **Time series and identification:** VAR, BVAR (Minnesota), VECM, and TVP-VAR; structural identification via Cholesky, long-run (Blanchard–Quah), sign and sign-zero restrictions, proxy/external instruments, and heteroskedasticity; impulse responses, variance decompositions, and bootstrap bands.
- **Single-equation and panel:** local projections (including IV and panel), dynamic-panel GMM, HAC and weak-instrument-robust inference, difference-in-differences, and synthetic control.
- **Computational macro:** value-function iteration, the endogenous grid method, Aiyagari and Huggett equilibria, and linear DSGE solution.
- **Reproducibility and teaching:** an in-browser JupyterLite playground and bilingual (English/Spanish) documentation and example notebooks.

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